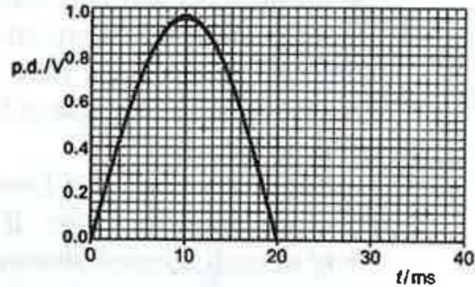


	0.2 m apart, there will be a node.
4(a)	From the graph, period $T = 40 \text{ ms}$, peak voltage $v_o = 4.8 \text{ V}$. $v = v_o \sin \omega t$, where $\omega = \frac{2\pi}{T}$ $v = 4.8 \sin\left(\frac{2\pi}{40 \times 10^{-3}}\right)t = 4.8 \sin(157t) \text{ V}$
4(b)(i)	$v = f\lambda$ $\lambda = \frac{340}{400} = 0.850 \text{ m}$ distance between adjacent nodes $= \frac{1}{2}\lambda = \frac{1}{2}(0.850) = 0.425 \approx 0.43 \text{ m}$
4(b)(i)	$R_{\text{total}} = 12 + \left(\frac{1}{6} + \frac{1}{6}\right)^{-1} = 15 \Omega$
4(b)(ii)	mean power $\langle P \rangle = \frac{V_{\text{r.m.s.}}^2}{R} = \frac{\left(\frac{4.8}{\sqrt{2}}\right)^2}{15} = 0.77 \text{ W}$

4(b)(iii)



Connecting a diode in series with a resistor will produce a half-wave rectifier and thus the p.d. across P will be only half a sine wave.

Using potential divider concept,

$$\text{peak p.d. across P} = \frac{4.8}{15} \times 3 = 0.96 \text{ V}$$

5(a)

u. I